

FINITE-SAMPLE SAFE MODEL RANKING UNDER SUBGROUP-MIX TURNOVER

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ABSTRACT

At deployment the class/subgroup mixture of a fixed pool of candidate models can shift (label-prior / subgroup-mix turnover) while each subgroup’s class-conditional risk stays unchanged, so a model optimal for the collected mixture may cease to be optimal—and a worse model can be selected. We formalize *safe ranking* as a finite-sample certification problem: given a calibration split satisfying the stated FIT/CAL sampling contract, decide for every future mixture weight w whether to *commit* a model with a certificate that its regret (loss minus the oracle-best model’s loss at w) is at most an operator-set tolerance τ , and otherwise *abstain* honestly rather than hard-select a possibly-overturning pick. We prove a linear-band structure (P1) giving joint coverage $\geq 1 - \delta$ of every mixture risk by weighted Clopper–Pearson bands, and a *paired-difference* (MCB-style) certificate (P3/Thm. 1) that bounds $\max\{0, \max_{j \neq i^*} (R_{i^*} - R_j)\}$ by pairing models on the *same* calibration points, cancelling shared error. On four public carriers and four classifiers, the dependent held-out estimated-regret diagnostic finds no committed row above τ (reported violation fraction 0/committed rows), and naively *minimizing the upper band* is shown to be a vacuous or *anti-selective* selector under heteroskedastic group widths—so we use *point-estimate selection + certificate gating* (M2/M2.5). The aggregate held-out diagnostic associates abstention with higher forced-point-estimate regret; it does not establish an exact abstention boundary or fallback utility. The primary empirical endpoint is the exact paired Maurer–Pontil (MPB) gate (0.260 committed; matched 5-seed mean); the normal 0.503 rate is a separately labelled asymptotic diagnostic, not finite-sample evidence. These historical endpoints are snapshot-reported only: E03 provides no application contract showing that its splits, sampling, or UCB implementation satisfied the stated finite-sample assumptions. M3/M3.5 are *oracle-stratified constructed-mixture diagnostics*: true class is unavailable before costly labeling absent an external stratified frame. Their aggregate rows neither specify a deployable allocation policy nor an acquisition study; the width surrogate does not optimize the actual candidate- and mixture-indexed regret UCB.

1 INTRODUCTION

Model selection is usually done once, on the mixture where data was collected. But at deployment the label/subgroup mix can move—a model trained on general web traffic now serves a skewed query stream; a classifier tuned on balanced data is deployed onto a single class-dominant population. Under *label-prior shift* (*subgroup-mix turnover*) the class-conditional distribution $P(X | Y=g)$ is fixed but the mixture weight w over subgroups is not. A model i ’s risk is then linear in w ,

$$R_i(w) = \sum_g w_g r_{i,g}, \quad r_{i,g} = \mathbb{E}[\ell(\text{pred}_i, Y) | Y=g], \quad (1)$$

so the identity of the best model, $\arg \min_i R_i(w)$, can change as w moves across the simplex. Standard selection heuristics live at one of two extremes: *single-point* selection (pick by the collected-mixture estimate) ignores w entirely; *worst-group* / *DRO* / *balanced-accuracy* selection ignores where w actually is. None answers the deployment question: *for a given future mixture w , can I both pick a model and certify its regret in finite sample—and when must I abstain rather than risk a pick that overturns $w \mapsto w'$?*

We study this as a *finite-sample safe ranking* problem with abstention: from a held-out calibration split, produce for each w a decision *commit*(model i^* and a certificate that its regret $\leq \tau$) or *abstain* (report that w is undecidable). The certificate is a finite-sample, coverage-guaranteed upper bound on the chosen model’s regret, and is allowed to abstain where the certificate is wider than the operator-set tolerance (Sec. 5, Tab. 2). Abstention is a certificate outcome, not an operational-utility claim: its fallback, latency, and monetary costs are unmeasured here.

1.1 CONTRIBUTIONS

1. **Conditional linear-band ranking structure** (Sec. 3): under the explicit FIT/CAL contract, if each entry $r_{i,g}$ is covered by a Clopper–Pearson interval with cell confidence $\delta/(Gk)$, then every mixture risk satisfies $L_i(w) \leq R_i(w) \leq U_i(w)$ with joint coverage $\geq 1 - \delta$ (P1, Lemma 1); best-vs-rest regret of any chosen model has a *sound* upper bound $U_{i^*}(w) - \min_j L_j(w)$ (P2).
2. **Paired-difference (MCB-style) certificate** (P3, Thm. 1): rather than subtract two *absolute* bands, bound $\max_j (R_{i^*} - R_j)$ by *paired within-CAL* differences (same points, same group). Pairing can reduce shared-error width, but no universal ordering against the absolute bound is claimed.
3. **A mechanism-level caution on “minimize-the-upper-band” selection** (Sec. 4): this pessimistic rule degenerates to the point estimate on homogeneous widths (vacuity) and *can* reverse rank opinion under heteroskedastic widths (Prop. 3); on real data it disagrees with point-estimate selection without a consistent direction. Framed as *bands gate, they do not choose*.
4. **Exact paired certificate-price frontier** (Sec. 5, Tab. 3): the primary exact paired Maurer–Pontil variant commits 0.260 vs. M2’s 0.269 (matched 5-seed) with reported held-out violation fraction 0 among committed rows. The 0.503 normal/CLT result is asymptotic only.
5. **Oracle-stratified diagnostic (M3, Sec. 6)**: historical FIT-specified oracle counts give descriptive aggregate contrasts, not executable label allocation; frozen summaries lack per-seed outcomes.
6. **Oracle width-surrogate diagnostic (M3.5, Sec. 7)**: for a stylized $\beta_g/\sqrt{n_g}$ width proxy, a single fixed mixture has the familiar water-filling solution $n_g \propto (w_g \beta_g)^{2/3}$ on positive coefficients. It only interprets same-budget oracle diagnostics, not the paired regret UCB or a deployment gate.

Label	Decision role	Status	Primary location
M1	upper-band selection	negative diagnostic	Sec. 4
M2/P2	estimate + absolute gate	exact certificate	Prop. 1, Tab. 2
M2.5/P3 (MPB/Hoeffding)	estimate + paired gate	conditional exact certificate	Thm. 1, Tab. 3
M2.5 normal	paired gate	asymptotic diagnostic	Tab. 3
M3	FIT-static oracle-stratum counts	constructed-mixture diagnostic	Sec. 6
M3.5	oracle width-surrogate counts	diagnostic, not a policy	Sec. 7

Table 1: Method-status taxonomy. “Exact certificate” is conditional on the stated turnover/calibration assumptions; every reported held-out quantity is a dependent diagnostic.

2 PROBLEM AND SETUP

Candidates. A fixed pool of k candidate models $\mathcal{M} = \{M_1, \dots, M_k\}$ is trained on a *FIT* split, disjoint from CAL and OUTER. Models are neither selected nor refit using CAL errors.

Subgroups. Data come from G subgroups (we use ground-truth classes). The deployment prior is a weight vector w on the simplex Δ^{G-1} ; per-model mixture risk is $R_i(w) = \sum_g w_g r_{i,g}$ ((1)).

Assumption 1 (subgroup-mix turnover). *During deployment only the mixture weight w shifts; each group’s class-conditional distribution $P(X \mid Y=g)$ is unchanged, so each $r_{i,g}$ is constant. Reward evaluation uses group-IPS reweighting.*

Assumption 2 (FIT/CAL sampling contract). *Conditional on the FIT data, the candidate pool, loss, subgroup definition, confidence allocation, and any CAL-count rule are fixed without reading CAL errors. FIT and CAL are independent. For each group g , the n_g CAL examples are conditionally i.i.d. from the unchanged group distribution $P_g = P(X, Y \mid Y = g)$. Thus, for fixed i , its CAL errors are i.i.d. Bernoulli with mean $r_{i,g}$; for fixed ordered (i, j) , the same-point transformed paired observations $X_{ij,g,t} = (d_{ij,g,t} + 1)/2 \in [0, 1]$ are conditionally i.i.d. OUTER is not used to choose, fit, or calibrate any certificate. Dependence among models on the same CAL point is allowed. Sampling without replacement, adaptive CAL choices based on CAL errors, or training/refitting on CAL lies outside this contract and needs its own valid finite-population or sequential analysis.*

Calibration. Under Assumption 2, CAL has n_g points in group g . We obtain $\hat{p}_{i,g}$ and a two-sided Clopper–Pearson interval $[l_{i,g}, u_{i,g}]$ with cell failure probability $\delta_{\text{cell}} = \delta/(Gk)$ (coverage at least $1 - \delta_{\text{cell}}$).

Decision problem. For each deployment mixture w , output either *commit*(i^* , certificate “true regret $\leq \tau$ ”) or *abstain*. Regret of a choice i at w is

$$\text{regret}(i, w) = R_i(w) - \min_j R_j(w),$$

i.e. the loss of choosing i versus the oracle-best model at w . We certify this quantity from calibration data *only*, never touching test.

3 THEORY: LINEAR BANDS AND THE REGRET CERTIFICATE

Throughout, $k = |\mathcal{M}|$, G subgroups, $\delta \in (0, 1)$, and Assumption 2 holds. Let $\delta_{\text{cell}} = \delta/(Gk)$, and let $[l_{i,g}, u_{i,g}]$ be the CP interval for $r_{i,g}$ with failure probability δ_{cell} . There are exactly kG absolute-band cells.

Lemma 1 (Linear band joint coverage). *Let E_{abs} be the event that $r_{i,g} \in [l_{i,g}, u_{i,g}]$ for all (i, g) . By the conditional i.i.d. Bernoulli condition in Assumption 2, each CP interval fails with probability at most δ_{cell} ; Bonferroni over the exactly kG cells gives $\Pr(E_{\text{abs}}) \geq 1 - \delta$. On E_{abs} , for every $w \in \Delta^{G-1}$ and every i ,*

$$L_i(w) := \sum_g w_g l_{i,g} \leq R_i(w) \leq \sum_g w_g u_{i,g} =: U_i(w).$$

Proof. Assumption 2 supplies the i.i.d. Bernoulli model required by the CP bound. Bonferroni over the Gk cells gives $\Pr(E_{\text{abs}}) \geq 1 - \delta$. On E_{abs} , fix w ; since $w_g \geq 0$, $\sum_g w_g = 1$, termwise monotonicity of the (deterministic) linear combination gives the sandwich. $\square$

Proposition 1 (Sound minimax regret bound (P2)). *Let i^* be any (possibly CAL-data-dependent) chosen model and define $\text{UB}_{i^*}(w) = U_{i^*}(w) - \min_j L_j(w)$. On E_{abs} ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{i^*}(w).$$

Proof. $R_{i^*}(w) \leq U_{i^*}(w)$ (Lemma 1), and $\min_j R_j(w) \geq \min_j L_j(w)$ since $R_j \geq L_j$ pointwise. Subtracting the two (weaker) inequalities at their respective endpoints yields the bound; this holds for the comparator j attaining the minimum. $\square$

Proposition 1 is *correct* but, as we show empirically (Sec. 4), unsuitable as a selector. It is a good *gate*: committing i^* whenever $\text{UB}_{i^*}(w) \leq \tau$ certifies true regret $\leq \tau$ with coverage $\geq 1 - \delta$. The tighter the bound, the more mixtures can be committed. This motivates the paired certificate, which is the central finite-sample construction.

Definition 1 (Paired-difference certificate). *For ordered pairs (i, j) , $i \neq j$, define the per-point paired difference on the calibration set*

$$d_{ij}(\mathbf{x}) = \mathbb{K}\{\text{err}_i(\mathbf{x})\} - \mathbb{K}\{\text{err}_j(\mathbf{x})\} \in \{-1, 0, +1\},$$

and its group- g mean $\bar{d}_{ij,g}$, an unbiased estimator of $r_{i,g} - r_{j,g}$. There are $m_{\text{pair}} = k(k-1)G$ ordered-pair/group cells. Set $\alpha = \delta/m_{\text{pair}}$ and, for every (i, j, g) , construct a one-sided upper bound $\text{UCB}_{ij,g}$ with marginal failure probability at most α . The paired Hoeffding and Maurer–Pontil (MPB) variants are used only for the conditionally i.i.d. bounded observations $X_{ij,g,t} = (d_{ij,g,t} + 1)/2 \in [0, 1]$ supplied by Assumption 2; the empirical-Bernstein/MPB variant additionally requires its formula’s $n_g \geq 2$ variance condition. Bonferroni over the exactly m_{pair} cells gives the simultaneous event E_{pair} with $\Pr(E_{\text{pair}}) \geq 1 - \delta$. The certificate value for chosen i^ at w is*

$$\text{UB}_{\text{cf}}(i^*, w) = \max\left\{0, \max_{j \neq i^*} \sum_g w_g \text{UCB}_{i^*j,g}\right\}.$$

Theorem 1 (Paired-difference certificate is sound (MCB-style)). *Under Assumption 2, on E_{pair} of Definition 1 (probability $\geq 1 - \delta$), for any, including CAL-data-dependent, choice i^* ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{\text{cf}}(i^*, w).$$

Hence if $\text{UB}_{\text{cf}}(i^, w) \leq \tau$, committing i^* is certified to have true regret $\leq \tau$, with coverage $\geq 1 - \delta$.*

Proof. $\text{regret}(i^*, w) = \max\{0, \max_{j \neq i^*} (R_{i^*}(w) - R_j(w))\}$: including the self-comparator contributes deterministically 0, and the minimum is attained by a comparator. If the inner maximum is positive, fix a maximizing $j_* \neq i^*$. Since E_{pair} holds, $R_{i^*}(w) - R_{j_*}(w) = \sum_g w_g (r_{i^*,g} - r_{j_*,g}) \leq \sum_g w_g \text{UCB}_{i^*j_*,g}$. This sum is exactly one term of the max in UB_{cf} , hence is bounded by it. If the inner maximum is nonpositive, regret is 0 and the explicit outer maximum supplies the required bound. $\square$

The key is *paired* cancellation: UB_{cf} subtracts models evaluated on the *same* calibration points within each group, removing the shared per-point error that can inflate UB_{i^*} in Proposition 1.

Honest scope of the certificate. Each $\text{UCB}_{ij,g}$ is computed once, independently of w (Bonferroni over the $k(k-1)G$ ordered-pair/group cells only), so the single event E_{pair} has probability $\geq 1 - \delta$ under Assumption 2 and $\text{UB}_{\text{cf}}(i^*, w) = \max\{0, \max_{j \neq i^*} \sum_g w_g \text{UCB}_{i^*j,g}\}$ is valid at the same time for *all* w on the simplex and all data-dependent choices $i^*(w)$: the paired certificate is *simultaneous over the continuous simplex*, with no grid-level Bonferroni. The historical snapshot also reports a dense off-grid Dirichlet w diagnostic (Sec. 7, 1000 points per (carrier, seed, budget), none on-grid): it finds no committed violation (e.g. MNIST $\pi=0.5$ commits 2816/3000 off-grid rows). The certificate certifies *soundness* of commits, not optimal abstention; τ is an operator parameter, and we report the coverage–commitment tradeoff as a fact, not a theorem. Crucially, E03 supplies only aggregate historical rows: it contains neither a split manifest nor CAL sampling/collection provenance, paired sufficient statistics, or the executed UCB code. It therefore cannot show that its historical split satisfied Assumption 2 or the CP/Hoeffding/MPB implementation conditions; its numerical rows remain snapshot-reported diagnostics rather than evidence of realized coverage.

4 WHY BANDS MUST GATE, NOT CHOOSE: M1 VERIFIED NEGATIVE

The most tempting “safety” rule is minimax selection,

$$i_{\text{mmp}}^* = \arg \min_i \left(\max_j \text{UB}_{ij}(w) \right), \quad \text{UB}_{ij}(w) = U_i(w) - L_j(w),$$

i.e. pick the model whose worst-case gap to any other is smallest. We ran it on real data (digits, Fashion-MNIST, MNIST, 20News; $G=\text{class}$; $k=4$; LR/C-variants, SVC, MLP).

Proposition 2 (M1 degeneracy). $\max_j (U_i(w) - L_j(w)) = U_i(w) - \min_j L_j(w)$, so $i_{\text{mmp}}^* = \arg \min_i U_i(w)$: the “max joint” collapses to the minimizer of the upper band, whose lower side ($\min_j L_j$) is constant across i .

When CP widths are homogeneous, $\arg \min_i U_i(w)$ equals the point-estimate argmin (vacuity); under *heteroskedastic* widths the rule can reverse a point-estimate ranking. On our real grid it matches point-estimate selection on every homogeneous-width row and diverges on mixture-ball vertices without a consistent direction (App. A). Thus bands are certificates, not selection arrows; the following two-model fact gives the precise possible failure mode.

Proposition 3 (Anti-selectivity of M1 under heteroskedastic widths). *For a fixed w and exactly two candidates a, b , suppose*

$$U_a(w) < U_b(w), \quad \hat{R}_b(w) < \hat{R}_a(w), \quad R_b(w) < R_a(w).$$

Then M1 selects a , whereas point-estimate selection selects b . The selected M1 regret is $\text{regret}(a, w) = R_a(w) - R_b(w) > 0$, while the point-estimate choice has zero regret. For example, $\hat{R}_b = 0.10 < 0.12 = \hat{R}_a$, $[L_a, U_a] = [0.02, 0.13]$, $[L_b, U_b] = [0.09, 0.20]$, and $R_b = 0.10 < R_a = 0.12$ meet every premise: M1 chooses a and has regret 0.02.

Proof. By Prop. 2, with two candidates M1 selects the smaller upper band, hence a . The point estimate selects b by the second premise. The third premise makes b oracle-best, so the definition in Sec. 2 gives $\text{regret}(a, w) = R_a(w) - R_b(w) > 0$. $\square$

Safety bands are *certificates*, not arrows; we adopt the design they motivate.

5 POINT-ESTIMATE SELECTION WITH CERTIFICATE GATING (M2 / M2.5)

5.1 THE VERIFIED POLICY

Selection is driven by the point estimate, never by the bands:

$$i^*(w) = \arg \min_i \hat{R}_i(w) = \arg \min_i \sum_g w_g \hat{p}_{i,g}.$$

Bands serve *only* as a gate for certificates and honest abstention. We commit $\langle i^*(w), \text{regret} \leq \tau \rangle$ iff $\text{UB}(i^*, w) \leq \tau$ (P2) or $\text{UB}_{\text{cf}}(i^*, w) \leq \tau$ (P3/Thm. 1), and otherwise abstain—falling back to a robust mixture / domain choice rather than a hard pick. This paper does not fix or evaluate that fallback policy or its operator cost; an abstention therefore means only “no certificate at tolerance τ ,” not “better deployment utility.”

Definition 2 (Cert-gated safe selection). *Given tolerance τ and a sound regret upper bound UB:*

$$\text{decision}(w) = \begin{cases} \text{commit } \langle i^*(w), \text{regret} \leq \tau \rangle & \text{UB}(i^*, w) \leq \tau, \\ \text{abstain} & \text{UB}(i^*, w) > \tau. \end{cases}$$

5.2 SNAPSHOT-REPORTED EXACT PAIRED FRONTIER (R1884 M2 + R1885 M2.5)

The historical snapshot reports four public carriers (digits, Fashion-MNIST, MNIST via PCA-128 frozen features; 20News via tf-idf→PCA-128), classes as subgroups, four classifiers (LR- $C \in \{1, 0.01\}$, LinearSVC, MLP), $\delta=0.1$, and a pre-registered w grid (uniform, collected u , 20 single-class skews, three interpolation peaks). Held-out estimated regret is settled on a test split by

group reweighting; test is only ever read. Rates below are *matched* on the same 5 seeds $\{0, 1, 2, 3, 4\}$ (Sec. 9). As explained after Theorem 1, E03 cannot verify that those historical splits meet Assumption 2; these are reported diagnostics, not an end-to-end replay.

Endpoint ($\tau=0.04$)	held-out pass frac.	commit rate	comm. mean/max	abst. mean	forced
M2 (P2, absolute bands)	1.000	0.269	0.0001/0.0019	0.0043	0.0031
M2.5 paired normal (asympt.)	1.000	0.503	0.0009/0.0364	0.0054	0.0031

Table 2: Snapshot-reported endpoints (*matched* 5-seed means, $\tau=0.04$). The primary exact paired MPB comparison is reported separately in Table 3. “1.000” means zero observed held-out estimated-regret violations among committed, dependent grid rows; it is not nominal coverage or a true-regret estimate. The M2 formula is conditional exact finite-sample; the paired-normal row is an asymptotic diagnostic. Exactness is conditional on Assumption 2, which E03 does not establish for these historical rows. Abstained rows have higher observed forced-pick regret (0.0044 vs. 0.00001), but this does not price a fallback. Same seeds/splits/pool.

Certificate	committed rate	held-out pass frac.	finite sample
M2 absolute CP bands	0.269	1.000	exact
M2.5 Hoeffding (paired)	0.097	1.000	exact
M2.5 Maurer–Pontil (paired)	0.260	1.000	exact
M2.5 normal/CLT (paired)	0.503	1.000	asymptotic

Table 3: **Exact paired certificate-price frontier** (snapshot-reported matched 5-seed means). The first three formulas are conditional exact finite-sample certificates; E03 provides no application contract showing that its historical rows met their assumptions. Their reported held-out violation fraction is 0. The normal/CLT row is an *asymptotic diagnostic*: its 0.503 rate and held-out diagnostic cannot support a finite-sample claim. The exact paired Maurer–Pontil bound commits about as many mixtures as M2 ($0.260 \approx 0.269$).

5.2.1 INTERPRETATION

1. **Exact versus asymptotic evidence.** Across the full $\tau \in \{0.02, \dots, 0.10\}$ sweep, the exact M2 and paired exact variants have no observed held-out violations in this snapshot. The normal/CLT variant also has no observed violations, but it is asymptotic and is not used as finite-sample evidence.
2. **Descriptive abstention pattern.** Abstained rows’ forced hard-pick regret is higher than committed rows’ (M2 matched: 0.0044 vs. 0.00001 mean). This descriptive contrast neither selects a fallback nor establishes its utility, latency, or cost.
3. **Per-carrier.** Large-error carriers (digits/20News) stay conservative and honestly abstain; low-error carriers (Fashion/MNIST) commit narrowly-and-safely. Reported heterogeneity, not hidden.
4. **Certificate-price, not rule change.** UB_{cf} cancels the within-CAL shared error that inflates UB_{i^*} in Proposition 1; the chosen i^* is identical, so the raise is honest safety-in-place.

6 ORACLE-STRATIFIED CONSTRUCTED-MIXTURE DIAGNOSTIC (M3)

M3 is an *oracle-stratified constructed-mixture diagnostic*, not label acquisition: it varies true-class counts in historical CAL arrays already separated by outcome. Before labeling, class is unobservable, so this would require an external frame whose membership is available independently of the costly label. E04 supplies neither such a frame nor collection/split provenance; M3 therefore neither specifies a deployable policy nor establishes Assumption 2 for its historical CAL sample.

6.1 ORACLE SETUP AND COUNT RULES

Fix $R = \lfloor \pi \cdot N_{\text{cal}} \rfloor$ with $\pi \in \{0.50, 0.65, 0.80, 0.95\}$. All other choices are inherited from M2.5 (point-estimate selection, paired-MPB gate, $\delta=0.1$, 3 seeds, four carriers). Three static rules map oracle inputs $(R, \{\text{per-group availability}\})$ to oracle stratum counts $\{n_g : \sum n_g = R\}$:

- **uniform**: $n_g \propto 1$ (equal across groups).
- **neyman / widthgreedy**: $n_g \propto \hat{\sigma}_g$ (per-group std of the FIT errors across the model pool), i.e. the error-band-width proxy.
- **sens**: $n_g \propto w_g^{\text{worst}} \cdot \hat{\sigma}_g$, where w^{worst} is the mixture vertex with maximal FIT-pool mean risk—sensitivity weighting.

The displayed count functions use FIT statistics, but FIT-only choice alone does not make the class-stratified collection feasible. Conditional on an external frame and conditionally i.i.d. within-frame draws that satisfy Assumption 2, such pre-fixed counts could be used by Thm. 1; E04 does not demonstrate those conditions. An earlier adaptive track and its claimed confidence-sequence result are omitted: this snapshot does not specify the observable subgroup information, filtration, sampling model, or low-count treatment needed for that theorem.

carrier	budget π	uniform	neyman	sens
MNIST	0.50	0.778	0.400	0.578
MNIST	0.95	1.000	1.000	1.000
Fashion	0.50	0.000	0.067	0.067
Fashion	0.95	0.133	0.111	0.133
digits	0.50~0.95	0.000	0.000	0.000
news	0.50~0.95	0.000	0.000	0.000

Table 4: **M3 oracle-stratified committed rate at $\tau=0.04$, $\delta=0.1$, 3-seed means.** Entries aggregate dependent constructed-mixture rows after true-class stratification. The frozen summary records no held-out estimated-regret violation among the 3360 oracle-rule cells, but this is neither nominal coverage nor evidence of feasible collection or an independent uncertainty estimate. No directional count-rule comparison is inferential here.

6.2 FINDINGS

1. **Conditional certificate scope.** An externally framed, conditionally i.i.d. stratified CAL design with FIT-fixed counts would fall under Thm. 1; the supplied M3 package does not prove that model. Its 0/3360 count is a dependent settlement diagnostic, not an estimate of nominal coverage.
2. **Oracle count contrasts are exploratory.** Uniform is 0.778 versus 0.400 (neyman/widthgreedy) and 0.578 (sens) on the MNIST $\pi=0.5$ summary; Fashion reverses the displayed ordering at low budget. Frozen M3 records contain only three-seed aggregate rows, not per-seed outcomes, so these comparisons are descriptive rather than supported directional conclusions.
3. **Missing defer costs.** We record forced hard-pick and FIT-anchor regret diagnostics (up to ~ 0.055 digits / ~ 0.048 news), but neither is a precommitted fallback policy and their utility, latency, and monetary costs are unmeasured. Details and the honest missing-data/sensitivity protocol are in App. D.

7 FIXED-BUDGET CERTIFICATE-WIDTH SURROGATE

This stylized *certificate-width surrogate* interprets M3’s oracle-stratified constructed-mixture contrasts only. It does **not** optimize the actual candidate-indexed paired regret UCB $\max_j \sum_g w_g \text{UCB}_{i^*j,g}$, identify an acquisition rule, or turn a commitment contrast into a deployment recommendation.

7.1 PROBLEM AND WATER-FILLING SOLUTION

Let the diagnostic total R be split into oracle stratum counts n_g , $\sum_g n_g = R$; true-class partitions are not available to an unlabeled-pool operator absent an external frame. Discarding candidate-specific empirical means, variances, bias terms, and comparator maxima, we retain $r_g(n) = \beta_g/\sqrt{n}$, where β_g is a FIT-visible dispersion proxy. For each fixed mixture w write $c_g(w) = w_g\beta_g$ and use the extended-value convention $c_g/\sqrt{0} = 0$ when $c_g = 0$ and $c_g/\sqrt{0} = +\infty$ when $c_g > 0$. The resulting objective is a width diagnostic, not a bound on regret or on any actual certificate:

$$\min_{n_g \geq 0, \sum_g n_g = R} \max_{w \in W} \sum_g w_g r_g(n_g). \quad (\text{WS})$$

Equation (WS) is the *multi-mixture* surrogate. The following closed form is only for its one-mixture subproblem $W = \{w\}$: with $A = \{g : c_g(w) > 0\}$, the continuous relaxation puts $n_g^* = 0$ for $g \notin A$ and, for $g \in A$, obeys equi-marginality:

$$\forall g \in A : \frac{1}{2}n_g^{-3/2}c_g(w) = \lambda \implies n_g^* = R \frac{c_g(w)^{2/3}}{\sum_{h \in A} c_h(w)^{2/3}}, \quad (2)$$

where λ is the budget multiplier. If $A = \emptyset$, the one-mixture surrogate objective is identically zero and every allocation is optimal. The $2/3$ exponent comes from the $\beta/\sqrt{n}$ scale (error-indicator concentration); it changes to $1/2$ or 1 for log or β/n costs. A bisection fill solves the budget equation in $O(G \log R)$.

Proposition 4 (Uniform optimality for one fixed width surrogate). *For a single fixed mixture $W = \{w\}$ with at least one positive coefficient $c_g = w_g\beta_g$, the all- G -group uniform allocation $n_g = R/G$ is the unique minimizer of the width surrogate (WS) iff $c_g = c > 0$ for every group g . If some $c_g = 0$ and some $c_h > 0$, the minimizer is on the boundary ($n_g^* = 0$ for zero-coefficient groups), so the all- G -group uniform allocation is strictly suboptimal. If every $c_g = 0$, the objective is identically zero and uniform is merely one of many minimizers.*

Sketch. When $A \neq \emptyset$, (2) is the unique optimum on its positive support. It equals R/G in every coordinate exactly when A is all groups and all c_g are the same strictly positive value. If A is a strict subset, moving any budget from $g \notin A$ to a positive-coefficient group strictly lowers the objective. The $A = \emptyset$ case follows from the convention above. $\square$

Multi-mixture surrogate calculation. For a finite predeclared $W = \{w_1, \dots, w_K\}$, the optimum is *not* obtained by applying the preceding formula to an arbitrarily chosen binding mixture. Writing each predeclared mixture as an engine with coefficient $C_{kg} = w_{k,g}\beta_g$, the surrogate uses the diagnostic dual:

$$\begin{aligned} d^* = P^* &= \min_n \max_{\mu \in \Delta^K} \sum_{k,g} \mu_k C_{kg} n_g^{-1/2} \\ &= R^{-1/2} \left[\max_{\mu \in \Delta^K} S(\mu) \right]^{3/2}, \quad S(\mu) = \sum_g \left(\sum_k \mu_k C_{kg} \right)^{2/3}, \end{aligned} \quad (3)$$

a convex diagnostic program (with S concave on Δ^K) solved by projected-gradient ascent. Complementarity identifies binding *surrogate mixtures*, not actual candidate comparators: 6–25% bind in these calculations, and the all-active heuristic changes the surrogate value by 1.17 – $4.19\times$ (App. B). On digits/news the surrogate stays above τ at every $\pi \leq 0.95$; this is a width-wall diagnostic, not infeasibility of the actual certificate.

7.2 DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

For the one-mixture surrogate, a deterministic construction (App. C) shows that uniform need not minimize stylized width: on an asymmetric non-spanning $W_0 = \{e_0\}$, three coefficients are zero, so continuous water-filling puts the *surrogate* budget on the queried group. This is a calculation about Eq. (WS), not a statement that the resulting allocation lowers the actual paired regret UCB or commits safely.

7.3 SAME-BUDGET ORACLE-STRATIFIED COMPARISON

We add a surrogate-water-fill oracle count assignment (with β_g from FIT-only cross-model paired-difference standard deviations) as a descriptive comparator. The historical paired-MPB values are reported for every oracle assignment; the surrogate neither upper-bounds them nor selects a safe allocation. Table 5 reports committed rate at $\tau=0.04$, 3-seed means.

carrier	budget	uniform	neyman	width	sens	width-surrogate
MNIST	0.50	0.778	0.400	0.400	0.578	0.400
MNIST	0.80	1.000	0.844	0.844	1.000	0.889
Fashion	0.50	0.000	0.067	0.067	0.067	0.067
Fashion	0.80	0.067	0.089	0.089	0.067	0.089
digits/news	all	0.000	0.000	0.000	0.000	0.000

Table 5: **M3.5 oracle width-surrogate comparator**, $\tau=0.04$, $\delta=0.1$, **3-seed means**. Same constructed-mixture grid and historical paired-certificate values as M3; the frozen dependent held-out diagnostic has no committed estimated-regret violation. These are descriptive oracle-stratified outcomes, not actual regret-UCB optimality or feasible label acquisition: uniform leads on MNIST, the surrogate ties neyman/width on low-budget Fashion, and digits/news never commit.

8 RELATED WORK

Group/DRO objectives (Sagawa et al., 2019; Duchi & Namkoong, 2019) optimize predictors, and prior adjustment (Saerens et al., 2002) recalibrates a chosen classifier; neither is this candidate-pool mixture-regret certificate. We instantiate CP/simultaneous-confidence ideas (Vovk et al., 2005; Clopper & Pearson, 1934), Maurer–Pontil empirical Bernstein (Maurer & Pontil, 2009), and paired comparisons (Dunnnett, 1955; Hsu, 1984) for that restricted decision object. M3 is only oracle-stratified diagnostic counting, not a label-allocation contribution.

9 LIMITATIONS

- **Finite-sample vs. asymptotic.** The paired certificate is simultaneous over the continuous simplex; the off-grid scan has no observed held-out violations. Only Maurer–Pontil/Hoeffding are finite-sample; normal/CLT is asymptotic. The exact relative gate has only trivial status-quo commits, so strict operators should use the absolute gate.
- **τ , dependence, and seeds.** We report a certificate-gate/commitment tradeoff, not an optimal- τ claim. M2/M2.5 use matched five seeds; M3/M3.5 use three-seed aggregate oracle-count means without retained per-seed records, so their directions are exploratory.
- **Oracle strata and replay.** Ground-truth class is not observable before costly labeling; absent an external stratified frame, M3/M3.5 are not acquisition procedures. E03/E04 lack split, sampling, sufficient-statistic, and executed-UCB provenance, so they cannot establish that the historical rows satisfy Assumption 2.
- **Turnover model.** Assumption 1 (fixed class-conditional) does not cover covariate shift inside a group; a distinct failure mode we do not certify.

10 CONCLUSION

Under Assumptions 1 and 2, exact CP and paired bounds certify point-estimate ranking through a regret gate. The exact paired MPB construction is the core result: its simultaneous event covers the continuous simplex, while its frozen 0.260 endpoint is only snapshot-reported until split and UCB provenance are released. Normal/CLT is asymptotic only. M3/M3.5 remain oracle-stratified constructed-mixture diagnostics, not label-acquisition or actual-regret-UCB policies; natural time/geo shift and fallback cost remain open.

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A M1 DIAGNOSTIC DETAIL

On the flat mixture grid, minimax-upper-band ($M1$) is identical to point-estimate selection on 350/350 rows (vacuity: homogeneous widths collapse $\arg \min_i U_i$ to the point-estimate $\arg \min$). On mixture-ball vertices that hedge over an uncertainty set it diverges on 165/352 rows but with no consistent direction on the diverged subset (mean regret diff -0.0021 : $M1$ slightly *better* on 93 rows, worse on 62), while over all 352 vertices it is slightly worse overall ($+0.0019$). On a diagnostic unbounded grid it diverges in 15/880 rows; those rows are mostly tied (11 of 15), the remaining 4 strictly worse for $M1$ (0.0024 vs. 0.0).

Setting	rows	diverge	regret diff
Flat grid (uniform/skew/interp)	350	0/350	$\equiv$ cal-prior
Mixture-ball	352	165/352	-0.0021 div. (93/62), $+0.0019$ all
Diagnostic grid	880	15/880	0 (11 tied; 4 worse for $M1$, diff $+0.0024$)

Table 6: Minimax-upper-band ($M1$) vs. point-estimate selection. “diverge” = rows where $M1$ ’s choice differs from cal-prior; regret diff = $M1$ regret minus cal-prior regret (positive = worse).

B ALL-ACTIVE WIDTH-SURROGATE ABLATION

For the multi-mixture width surrogate only, complementary slackness identifies a small fraction of binding engines. The dual value is not the actual candidate/mix regret UCB; the all-active column reports a ratio between two surrogate calculations.

carrier	$ K $	active engines	$d_{\text{mm}}^*/d_{\text{allactive}}^*$
MNIST	60	25%	1.30
Fashion	60	25%	1.33
digits	60	7%	4.19
news	100	20%	1.17

Table 7: All-active heuristic ablation for the width surrogate (Eq. 3), not for the actual paired-regret UCB.

C DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

Take $G=4$, $R=1200$, per-group paired-error std $\beta = (0.10, 0.10, 0.02, 0.02)$, true per-group gaps $\delta = (0.02, 0, 0.02, 0)$, and an *asymmetric* (non-spanning) deployed set $W_0 = \{e_0\}$ (only subgroup 0 is ever queried):

rule	n_g	reported paired UCB at e_0	actual commit
uniform	(300, 300, 300, 300)	0.094	no
width-surrogate	(1197, 1, 1, 1)	0.029	yes

Here $c = (0.10, 0, 0, 0)$, so Prop. 4 gives the continuous surrogate optimum $(R, 0, 0, 0)$; uniform is strictly suboptimal because it wastes labels on the three zero-coefficient groups. The displayed integer implementation enforces one label per inactive group and therefore uses (1197, 1, 1, 1), a near-boundary realization. Its reported paired UCB and “actual commit” are separate paired-certificate outputs, not numerical evaluations of Eq. (WS) and not an allocation guarantee for the actual UCB.

D DEFER COSTS IN M3

The snapshot records forced point-estimate and FIT-anchor regret diagnostics, but neither fallback was precommitted as an operational policy and no latency, monetary, switch, or abstention penalty

was measured. Thus the reported hard-pick regret (up to ~ 0.055 digits / ~ 0.048 news) is not a fallback-cost estimate. The planned real-shift protocol records these fields as missing or conducts an explicit sensitivity analysis; no utility conclusion is made here.

E GATE OPERATING CHARACTERISTIC

deployed set	$CV(\hat{\beta})$	signed area = $\text{mean}_{\tau}(\text{rate}_{\text{surrogate}} - \text{rate}_{\text{unif}})$	
		area	worst τ gap
non-spanning W'	low (sym.)	+0.25	—
non-spanning W'	high	+0.25	—
spanning W	low (sym.)	+0.00	0
spanning W	high	-0.08	-0.49 at $\tau=0.12$

Table 8: Diagnostic operating characteristic: committed-rate difference width-surrogate – uniform vs. τ ($G=4$, $R=1200$, 20 seeds, same paired-MPB certificate). This describes a synthetic comparison only; it is not a safe-allocation gate or an optimization result for the actual regret UCB. Evidence: `results/SUBGMIX_GATE_OC_R1899.json`.

F CHOOSING OR ELIMINATING THE TOLERANCE τ

The certificate band is τ -agnostic: τ enters only the comparison $D(i^*, w) \leq \tau$, never its construction (Sec. 3, Prop. 1). We report two distinct τ -related diagnostics. *M6* is an *asymptotic/descriptive relative-gate diagnostic*; its exact-relative counterpart is status-quo-only on the stored rows, so it is not a finite-sample deployment rule. *M7* retains the absolute gate. The primary finite-sample operational statement remains the conditional absolute *M2.5* certificate of Theorem 1, not either historical diagnostic. *M7* runs out-of-distribution from the main evidence on an unseen seed set $\{5, 6, 7, 8\}$ (the main tables use seeds $\{0, \dots, 4\}$); *M6* runs on the same seeds $\{0, \dots, 4\}$ as the main tables, and we additionally report its strictly-finite-sample band behavior below.

M6: asymptotic/descriptive relative-gate diagnostic. For a stored diagnostic comparison, replace the absolute oracle comparator by the *status-quo model the operator would have used anyway*: F_0 = point-estimate-optimal collected-mixture model (no certificate needed to run it). The normal-band diagnostic proposes the selected $i^*(w)$ only if its one-sided paired upper band is not worse than F_0 , else keeps F_0 . This is a *relative diagnostic* (no worse than the stored status quo), and τ never appears. Across all $350 \text{ rows} \times 4 \text{ carriers}$, the upgrade rate is 0.663; the squared regret difference $R_{\text{dec}} - R_{F_0}$ has mean -0.0015 and *maximum* 0.0, i.e. no-worse-than- F_0 held-out diagnostic violation fraction 0. On the 118 rows where *M6* abstains, running the naive always-commit rule instead would make 23 (19%) genuinely worse (max additive regret $+0.028$); *M6*’s max regret is 0 there. These are snapshot-reported normal-band diagnostics, not a finite-sample certificate or a safe-upgrade finding. The exact-relative check below fixes the stored F_0 and i^* and has no genuine switch: it only keeps the status quo. Thus an operator requiring the conditional finite-sample guarantee uses the absolute *M2.5* gate under Assumption 2; E03 does not establish that its historical rows meet that application contract.

M7: keep absolute τ , select it from a finite menu on calibration only. Take a finite menu $T = \{0.01, \dots, 0.05\}$ and the prespecified rule $\hat{\tau} = \min\{\tau \in T : \text{CR}_{\text{cal}}(\tau) \geq 0.5\}$ (else the tightest menu entry), using only the calibration block’s group errors — never the outer/test block. Because the band holds simultaneously independent of τ , the resulting $\hat{\tau}$ carries *zero additional certificate multiplicity*: every committed point still satisfies true regret $\leq \hat{\tau}$ on the same $\geq 1 - \delta$ event. The price of selection is confined to the performance layer. On OUTER settlement, CAL-select yields committed rate 0.357 with no observed held-out violation (mean committed regret 0.0003, max 0.0053), picking conservatively tight menu entries ($\hat{\tau}=0.01$ on 12 of 16 carrier $\times$ seed blocks). A test-snooping variant that picks τ on OUTER and reports the same test inflates committed rate to 0.539 ($+0.18$) with a 0.7% held-out estimated-regret violation rate — the precise price of the data-peeking the CAL-only rule avoids. Weak domains are reported unfiltered: digits commits 0/60 rows (capacity wall) and news 4/100 (budget wall).

Exact relative-gate status-quo check (M6 continued). The main M6 finding uses the *asymptotic* paired normal band. Holding the stored F_0 , i^* , paired differences, and Bonferroni split fixed, we also substitute the exact Hoeffding and Maurer–Pontil widths. This is a snapshot-reported fixed-selection diagnostic, not evidence that E03 met Assumption 2. Its result is a *vacuity boundary*: all 226 rows with $i^*=F_0$ commit trivially (difference 0), while the only six rows proposing $i^* \neq F_0$ are rejected by the exact widths (empirical paired difference $D \in [-0.036, 0.098]$, width ~ 0.25). Thus no inspected exact-relative commit differs from the status quo. The exact absolute M2.5 gate is the primary conditional finite-sample rule and retains snapshot-reported content (0.260 MPB committed, with zero observed held-out estimated-regret violations).

A separate width-only thought experiment keeps those six empirical paired differences and their F_0/i^* choices fixed while scaling only $n_g \rightarrow Nn_g$. In that fixed-statistic calculation, the six proposals reopen under exact MPB at $N^* \in \{2, \dots, 10\}$ (median 5) and Hoeffding at $N^* \in \{2, \dots, 20\}$ (median 10); their stored oracle switch-gain fields are 0.034–0.130. This algebra does not predict the outcome of collecting more data, because new data could change the empirical differences, F_0 , i^* , and the decision margin.

M9/M10 (audited): fixed-statistic required-budget calculation. E07 stores finite budget-grid rows generated from fresh seeds $\{10, \dots, 14\}$. Those rows are snapshot-reported observations; because a new CAL draw can change empirical paired means, F_0 , i^* , and the selector margin, they do not justify a claim beyond their inspected schedules.

For a separate algebraic calculation, fix one full-split row’s empirical paired means $\bar{d}_{i^*F_0,g}$, its selected F_0 and i^* , and $\Delta(w) := \hat{R}_{F_0}(w) - \hat{R}_{i^*}(w) = -\sum_g w_g \bar{d}_{i^*F_0,g} \geq 0$. Holding all of these statistics fixed, write the one-sided paired formula as

$$\text{UCB}_g(n_g) = \bar{d}_{i^*F_0,g} + \text{bw}_g(n_g), \quad \text{bw}_g(n_g) \geq 0.$$

Then its relative admission inequality is the deterministic identity

$$\sum_g w_g \text{UCB}_g(n_g) = -\Delta(w) + \sum_g w_g \text{bw}_g(n_g) \leq 0 \iff \sum_g w_g \text{bw}_g(n_g) \leq \Delta(w).$$

It does not invoke a coverage relation to an unknown mean. For the fixed-statistic Hoeffding formula $\text{bw}_g = c/\sqrt{n_g}$ and proportional counts $n_g = b n_g^{\text{full}}$, a necessary budget is $b \geq b^*(w) := [c \sum_g w_g / \sqrt{n_g^{\text{full}} / \Delta(w)}]^2$. For all 125 stored full-split real-switch rows ($i^* \neq F_0$), this *fixed-statistic* calculation has $b^* > 1$ (median 273, minimum 3.7; Δ median 0.014). Thus its proportional width counterfactual rejects at $b \leq 1$. Moreover, within the same counterfactual only, $n_g \leq n_g^{\text{full}}$ makes the Hoeffding width mass no smaller than at the full caps. Hence no capped allocation admits that *fixed* row under this formula. This monotonicity says nothing about new empirical means, a new selector/margin, or the nonmonotone small-sample MPB bias term.

Separately, the E07 stored grid reports zero genuine exact-relative switches at its inspected budget schedules. That is a finite-grid diagnostic, not a general allocation result. Trivial status-quo rows ($i^*=F_0$) are excluded from real-switch rates. The absolute M2.5 rows separately retain snapshot-reported content at these schedules, but become a finite-sample deployment certificate only if Assumption 2 is established. The relative normal gate remains asymptotic/descriptive.

M12 (historical oracle-stratified check at the M3/M3.5 totals). The frozen M12 artifact, not a field collection protocol, fixes a per-group oracle count $n_g = b n_g^{\text{full}}$ on fresh seeds $\{10, \dots, 14\}$ and reports the same object at the M3/M3.5 *total* budgets $R = \lfloor \pi N_{\text{cal}} \rfloor$, $\pi \in \{0.5, 0.65, 0.8, 0.95\}$, under reported without-replacement *FIT/CAL/OUTER* sampling (same four carriers, same pool, same $\delta, d_{\text{cell}}, w$ -grid, paired bands), re-splitting R among subgroups with the pre-specified oracle M3 count rules (uniform/neyman/sens/width-greedy) and the M3.5 width-surrogate water-fill. These true-class strata and without-replacement draws do not establish Assumption 2; the artifact contains no collection/split trace proving a valid alternative finite-population contract. Across all 100 cells (4 carriers $\times$ 4 budgets $\times$ 5 allocations $\times$ 5 fresh seeds), the exact *relative* band (Hoeffding or MPB) admits 0 **genuine switches** ($i^* \neq F_0$ committed): every reported exact-relative “commit” is a trivial keep-status-quo row. This is a finite stored-grid observation after the artifact’s own sampling and selection, not a conclusion beyond those cells, and it is not an extension of the fixed-statistic M9 calculation. The exact *absolute* gate (M2.5) snapshot reports no held-out estimated-regret violation on the 37 committing cells, with content monotone in the budget (MNIST uniform

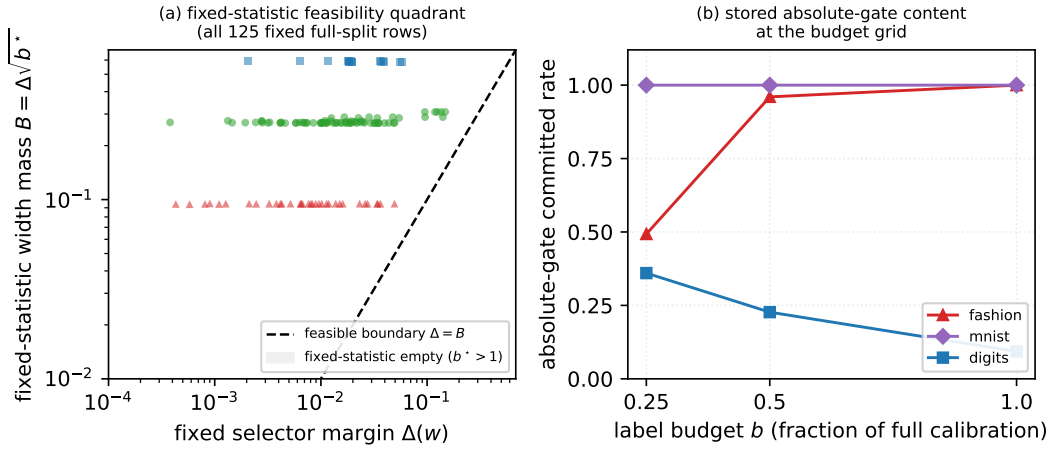


Figure 1: **Fixed-statistic required-width calculation and stored budget diagnostics.** (a) The 125 real-switch rows ($i^* \neq F_0$) at the full split across the fresh seeds $\{10, \dots, 14\}$, in the *selector-margin* $\Delta(w)$ vs. *required bandwidth-mass* $B = \Delta \sqrt{b^*(w)}$ quadrant (log-log; marker keyed by carrier). Admission under any fixed-statistic Hoeffding formula forces $\Delta(w) \geq B$ (the feasible half-plane below the dashed $\Delta=B$ line); every displayed row has $b^*(w) > 1$. The calculation fixes the empirical paired means, F_0 , i^* , and $\Delta(w)$ while varying only its width, so it does not cover a fresh CAL sample or a changed data-derived selector. (b) The conditional *absolute* exact gate (M2.5) instead retains snapshot-reported content monotonically in the label budget b : committed rate grows with b on fashion/digits while mnist commits at every budget, all with zero observed held-out no-worse-than- F_0 violations (Sec. F). The relative gate (M6) is an asymptotic/descriptive diagnostic only.

0.773 $\rightarrow$ 1.0; Fashion 0 $\rightarrow$ 0.20 at $\pi=0.95$, switch gain mean 0.0131/max 0.0753, worst harm -0.0012), and the weak domains keep their walls (digits/news 0 committed everywhere). These are historical oracle diagnostics, not coverage evidence or deployable allocation evidence. Evidence: subgroup_mix_ranking/results/SUBGMIX_M12_FRESH5_BUDGET_R1915.json.

G AI USE STATEMENT

Large language models, including Codex, assisted with implementation support, proof and cross-reference auditing, bibliography and metadata checking, manuscript revision, and internal review workflow support. They did not generate experimental observations, choose unreported results, or replace author responsibility for scientific claims, experiment design, interpretation, or the final manuscript. This statement includes the present revision and audit cycle.